From Reads to Rare Disease Diagnosis: Leveraging Trio-Based Variant Calling for Mendelian Disease Prediction

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ABSTRACT - Genomic analysis has become a cornerstone of disease diagnostics, enabling the detection of genetic mutations such as Single Nucleotide Polymorphisms (SNPs), small insertions and deletions (indels), and frameshift variants, which can alter gene function and contribute to the development of various diseases.
In this report, we analyzed chromosome 20 exomes from ten family trios obtained through Whole Exome Sequencing, with the aim of identifying candidate variants associated with rare Mendelian disorders. Initially, several quality control steps were performed on the raw dataset, including an assessment of read coverage.
Subsequently, variant calling was carried out to identify variants in each family member relative to the reference genome GRCh38.
The analysis focused on specific target regions of chromosome 20 known to be associated with diseases. Guided by the observed phenotypes (affected vs. unaffected status), different inheritance models were evaluated for each trio, including Autosomal Dominant, Autosomal Recessive, and De Novo patterns. The project was conducted using key computational tools, including a UNIX-based pipeline and ENSEMBL's Variant Effect Predictor. Our findings highlight the importance of integrating computational pipelines into the diagnostic workflow to enable the timely identification of causative mutations in rare genetic diseases.

1. Introduction

Genomic analysis is a transformative tool in modern medicine, facilitating the diagnosis of rare diseases that would otherwise remain elusive and time-consuming to identify. Furthermore, it paves the way for precision medicine by enabling targeted treatments and allowing for accurate calculation of recurrence risks for future offspring.

Rare genetic diseases are conditions caused by DNA alterations that, while affecting a small percentage of the population individually, represent approximately 80% of all rare conditions. A disease is generally defined as rare when it affects fewer than 1 in 2,500 individuals (global average). Although each specific disease is rare, the aggregate number of people living with these conditions is substantial, as thousands of different types have been documented.

These rare genetic disorders are typically classified based on their underlying etiology:

- Monogenic: Caused by a mutation in a single gene.
- Chromosomal: Resulting from changes in the structure or number of chromosomes.
- Multifactorial: Arising from a combination of genetic factors and environmental triggers.

This assay focuses on the analysis of rare monogenic variants that follow Mendelian laws of inheritance. Specifically, we consider the primary inheritance patterns that result in an affected child:

- Autosomal Dominant (AD): A single copy of the mutated gene (inherited from one parent or arising de novo) is sufficient to manifest the disease.
- Autosomal Recessive (AR): Two copies of the mutated gene (one from each parent) are required for the phenotype to be expressed.

We also explore the occurrence of de novo mutations. These arise in the father's sperm or the mother's egg; consequently, the mutation is absent

from the somatic cells of the parents, but is present in every cell of the offspring. Every individual is born with approximately 70 to 100 de novo germline mutations, subsequently accumulating thousands of somatic mutations throughout their lifetime. However, the vast majority of these variants are functionally neutral, as they predominantly occur within non-coding or non-functional regions of the genome.

It is statistically infrequent for a mutation to occur within a critical gene in a manner that fundamentally disrupts its biological function.

This report focuses on identifying rare autosomal disorders caused by the disruption of single-gene activity following a Mendelian model. Specifically, we analyzed a cohort consisting of five individuals from two distinct family trios. Genomic data were obtained via Whole Exome Sequencing (WES), a high-throughput technique that selectively targets the protein-coding regions (exons) of the genome. While the exome constitutes only about 1% to 2% of the entire human genome, it harbors approximately 85% of known disease-causing mutations. Consequently, exome sequencing is a highly efficient and cost-effective tool for clinical diagnostics and the identification of potentially pathogenic inherited variants.

The available clinical and genetic context for each trio is summarized below:

Trio Number	Mode	Notes
1	AR	NA
2	AD inherited	father affected
3	AD De Novo	NA
4	AR	NA
5	AR	NA

Fig. 1. AR stands for Autosomal Recessive. AD for Autosomal Dominant

For this analysis, the primary area of interest is Chromosome 20. Alterations in this chromosome are linked to a spectrum of disorders, including *Alagille Syndrome*, *Autosomal Dominant Sleep-Related Hypermotor Epilepsy*, and *MODY 1 (Maturity-Onset Diabetes of the Young, Type 1)*. Using bioinformatics tools, we investigated the relationship between specific genetic mutations and these rare disorders, accounting for the influence patterns of familial inheritance.

2. Materials and Methods

The study involves the analysis of two distinct families to investigate the presence of rare genetic disorders in children. Each family is divided into five specific cases, representing different Mendelian inheritance patterns to test. The primary dataset for each of the ten analyzed cases includes:

- Raw Sequencing Data: Two zipped FastQ files for each family member (Mother, Father, Child) obtained by paired-end sequencing. These contained both strands of the sequenced DNA.
- Reference Genome: The human reference genome (Chromosome 20, GRCh38)
- Disease Panel and Inheritance Models: A comprehensive list of target diseases and their expected inheritance modes, including Autosomal Recessive (AR), Autosomal Dominant (AD) inherited from the father, and De Novo mutations.

All of these files were shared on university server in a public directory. The connection via terminal to access the files was done using the Secure Shell (SHH) protocol. The commands used to access or retrieve files from the server will be omitted as they are outside the scope of this analysis.

A dedicated directory structure was created within the user's home environment to ensure a streamlined workflow. Each case was assigned a unique directory containing all requisite files. To optimize storage in large-scale analyses, symbolic links would typically be utilized for the reference genome; however, for the scope of this project, copies were maintained in each directory.

The file structure (visualized via ls -R) for the initial trio_1 of Cenzato is presented in the logs below (subsequent trios followed an identical organization).

```
.:
trio_1  trio_2  trio_3  trio_4  trio_5

./trio_1:
chr20.1.bt2
chr20.fa
chr20.rev.2.bt2
HG00428.targets_R2.fq.gz
chr20.2.bt2
chr20.fa.fai
HG00427.targets_R1.fq.gz
HG00429.targets_R1.fq.gz
chr20.3.bt2
chr20_ILMN_Exome_2.0_Plus_Panel.
   hg38_padded.bed
HG00427.targets_R2.fq.gz
```

```
HG00429.targets_R2.fq.gz
chr20.4.bt2  chr20.rev.1.bt2
HG00428.targets_R1.fq.gz
```

Each case were then treated independently by writing a custom script that includes all the steps that will be discussed in this section. The code snippets presented hereafter are derived from the analysis of the first trio; however, any case-specific deviations or modifications applied to subsequent trios will be explicitly noted.

Initially, the pipeline focused on the alignment and indexing of the raw reads. Alignment was performed using Bowtie2 aligner tool. To maximize computational efficiency and minimize the creation of large intermediate files, the output was piped directly into Samtools. The resulting Sequence Alignment Map (SAM) was converted to a Binary Alignment Map (BAM), that is then sorted and indexed for faster analysis.

The following command illustrates the processing of the child's sequence: Bowtie2 alignment tool and passes it to Samtools. The result of the alignment is a binary file in BAM format that is then indexed for faster analysis. This is the code for the procedure for the child sequence.

```
bowtie2 -1 HG00427.targets_R1.fq.gz -2
    HG00427.targets_R2.fq.gz  \
--rg-id child --rg "SM:child" \
-x chr20 \
-p 12 |
samtools view -Sb |
samtools sort -o child.bam

samtools index child.bam
```

To provide a brief technical overview of the alignment process: the Bowtie2 tool utilizes the paired-end sequencing reads (forward and reverse strands, specified by the -1 and -2 flags) and assigns a unique for them (child). The -x parameter directs the tool to the pre-indexed reference genome located in the working directory, thereby streamlining the command execution.

The resulting alignment is then channeled through a UNIX pipe to Samtools, which converts the data into a compressed binary format (BAM) and performs coordinate-based sorting. Finally, the BAM file is indexed to optimize performance for downstream analysis. An identical procedure was applied to the parental samples, ensuring each was assigned a distinct identifier (mother and father).

Before proceeding with variant searching, the quality of the sequenced data needed to be evaluated. Read quality was assessed via FastQC, while alignment metrics and coverage depth across the target exome panel (.bed file) were generated using Qualimap. Finally, MultiQC was employed to aggregate these individual reports into a single, comprehensive visualization.

```
fastqc child.bam
fastqc mother.bam
fastqc father.bam

qualimap bamqc -bam child.bam --feature
    -file chr20_ILMN_Exome_2.0
    _Plus_Panel.hg38_padded.bed \
--outdir child

qualimap bamqc -bam mother.bam --
    feature-file chr20_ILMN_Exome_2.0
    _Plus_Panel.hg38_padded.bed \
--outdir  mother

qualimap bamqc -bam father.bam --
    feature-file chr20_ILMN_Exome_2.0
    _Plus_Panel.hg38_padded.bed \
--outdir father

multiqc -f .
```

To facilitate visual inspection of the coverage tracks in the **Results** section, Bedtools was utilized to generate BedGraph files:

```
bedtools genomecov -ibam child.bam -bg
    -trackline -trackopts \
'name="child"' -max 100 \
> childCov.bg
```

Subsequently Variant calling was executed using FreeBayes tool. Specific quality thresholds were applied to ensure high-confidence calls:

- "-m" and "-q": Minimum mapping and base quality
- "-C": Minimum count of observation to support an alternative allele
- "-Q":Threshold quality over which a mismatch is counted

The resulting multi-sample VCF file was then compressed and indexed for further analysis.

```
freebayes -f chr20.fa \
-m 20 -C 5 -Q 10 -q 10 \
```

```
--min-coverage 10 \
child.bam father.bam mother.bam \
> multi_sample_trio_1.vcf

bgzip multi_sample_trio_1.vcf

bcftools index multi_sample_trio_1.vcf.gz
```

It is important to emphasize the specific order in which the BAM files were passed to the command. Since the downstream inference of inheritance patterns relies on sample indexing, a text file (hereafter referred to as samples.txt) was created. This file contains the unique ID's for each family member in a sequence consistent with the multi-sample VCF, ensuring accurate genotype attribution during the filtering stages.

```
child
father
mother
```

Variant filtering was performed using Bcftools, targeting the genomic regions specified in the exome panel BED file. The next sections it will be explained how each case was handled depending on the pattern of inheritance.

A. Autosomal Recessive Cases (1,4,5)

Rare autosomal recessive disorders typically manifest when a child inherits two dysfunctional alleles (one from each parent), resulting in a homozygous genotype. In this model, both parents are obligate carriers, meaning they are clinically asymptomatic and possess a heterozygous genotype at the target locus.

```
bcftools view -R chr20_ILMN_Exome_2.0_Plus_Panel.hg38_padded.bed multi_sample_trio_1.vcf.gz \
| bcftools view -S samples.txt \
| bcftools view -i 'GT[0]="AA" && GT[1]="RA" && GT[2]="RA"' \
| bcftools filter -i 'QUAL>20' -Ov -o multi_sample_trio_1.cand.vcf
```

A UNIX pipe is again employed to restrict the analysis to the genomic intervals specified in the BED file, followed by a sample-specific filtering process. The logical expression in the third line facilitates the selection of variants based on their allelic states: in Bcftools syntax, 'R' represents the reference allele, while 'A' denotes the alternative allele.

The samples are accessed via positional indexing (0 to 2), corresponding to the order established during the variant calling with FreeBayes (Child, Father, and Mother, respectively). In the final step, a quality threshold is applied to retain only high-confidence variants, which are then exported to a separate VCF file for downstream interpretation.

B. Autosomal Dominant Cases: inherited from the father (2)

The filtering procedure follows the previous logic, focusing on the paternal transmission of a dominant allele to a heterozygous child. While the father is typically expected to be heterozygous, the filter also accounts for homozygous dominant paternal genotypes to ensure comprehensive variant detection. Conversely, the maternal genotype must remain homozygous for the reference allele, indicating the absence of the variant in the mother.

```
bcftools view -R chr20_ILMN_Exome_2.0_Plus_Panel.hg38_padded.bed multi_sample_trio_2.vcf.gz \
| bcftools view -S samples.txt \
| bcftools view -i 'GT[0]="RA" && (GT[1]="RA" || GT[1]="AA") && GT[2]="RR"' \
| bcftools filter -i 'QUAL>20' -Ov -o multi_sample_trio_2.cand.vcf
```

C. Autosomal Dominant cases: De Novo (3)

De novo disorders arise from mutations occurring within the parental gametes (germline mutations). Consequently, the pathogenic allele is absent from the parents' genotype, whereas the child manifests at least a heterozygous state (RA), sufficient to express the phenotype in a dominant inheritance model.

```
bcftools view -R chr20_ILMN_Exome_2.0_Plus_Panel.hg38_padded.bed multi_sample_trio_3.vcf.gz \
| bcftools view -S samples.txt \
| bcftools view -i '(GT[0]="RA"||GT[0]="AA") && GT[1]="RR" && GT[2]="RR"' \
| bcftools filter -i 'QUAL>20' -Ov -o multi_sample_trio_3.cand.vcf
```

In order to identify the most likely causal variant, functional annotation was performed. Variants were filtered using the Ensembl Variant Effect Predictor (VEP) based on two critical criteria:

Predicted Impact: Only variants with a "HIGH" impact were retained.

Population Frequency: To prioritize rare diseases, variants with a Maximum Allele Frequency (MAX AF) greater than 0.0001 were excluded.

```
filter_vep -i multi_sample_trio_1.
   vep_annotated.vcf -o trio_1.
   vep_filtered.vcf \
--filter "IMPACT is HIGH and (not
   MAX_AF or MAX_AF < 0.0001)"
```

The filtered VCF files allow for the immediate identification of transcripts that exhibit the expected pathogenic behavior. To ensure the reliability of these findings, a cross-validation was performed using the *Ensembl Variant Effect Predictor* (VEP) web interface. By applying identical filtering criteria, targeting variants with a HIGH predicted impact, enabling the *Show Phenotypes* and choosing RefSeq annotation, we were able to corroborate the candidate variants and evaluate their known clinical associations.

3. Results

Results of the variant prioritization process are presented in **Figures 2 and 3**. Pathogenic Mendelian mutations were confirmed only when the identified genotype was strictly correlated with the clinical presentation and the expected mode of inheritance. In such instances, the tables specify the affected gene, the precise variant location, and the associated phenotypic traits. Cases yielding no candidates after stringent filtering are classified as negative for the targeted rare disorders.

A. Quality and Coverage assessment

Before genomic analysis, it was essential to evaluate the quality of the sequencing data to ensure the reliability of the downstream results. The sequencing process involves DNA fragmentation and base-calling, procedures that are inherently prone to stochastic errors. To address this, we performed quality control assessments and aligned the reads with the reference genome GRCh38 using Bowtie2 to accurately reconstruct the original genomic sequences. A fundamental metric in this assessment is coverage, which represents the number of unique sequencing reads that align to a specific base or region. We distinguish between:

- Coverage Depth: The average number of times a single base is sequenced.
- Coverage Breadth: The percentage of the target genome covered by at least a predefined number of reads.

Since modern sequencing protocols require DNA fragmentation, it is essential to achieve sufficient coverage depth—the number of times a specific nucleotide is sequenced—to distinguish true biological variants from technical artifacts. As a general benchmark, a minimum depth of 10X (ten independent reads per position) is required to ensure data reliability. In this analysis, we defined our quality standards based on the percentage of the genome meeting specific depth thresholds:

- Acceptable: 95% of the genome at > 10X.
- Good: 95% of the genome at > 20X.
- Excellent: 98% of the genome at > 20X

Since our initial DNA sequence data sets were stored as ASCII characters in FASTQ format, we utilized the associated quality scores to estimate the confidence level of each base call. These scores are encoded in a compact Phred quality format, a logarithmic metric used in DNA sequencing to quantify the probability of an incorrect base call. A higher score reflects greater confidence; for example, a Q30 score represents 99.9% accuracy (or 1 error in 1,000 bases).

Subsequently, due to the presence of repetitive regions in the human genome, we also calculated the MAPQ score(Mapping Quality). This Phred-scaled value, found in the SAM/BAM files, indicates

Cases Chiara	diagnosis	chr	Mutated gen	Variant Location	ref	alt	Impact	Consequence	Associated phenotype
Case 1	healthy	NA	NA	NA	NA	NA	NA	NA	NA
Case 2	pathogenic	20	SALL4	20:51788938	C	A	High	stop gained	DUANE-RADIAL RAY SYNDROME
Case 3	pathogenic	20	ASXL1	20:32428324	G	A	High	splice acceptor variant	Bohring-Opitz syndrome
Case 4	pathogenic	20	FERMT1	20:6097544-6097546	ACA	AA	High	frameshift_variant	FERMT1-related disorder|Kindler syndrome,
Case 5	pathogenic	20	ADA	20:44621016-44621016	A	G	High	splice_donor_variant	Severe combined immunodeficiency, autosomal recessive,T cell-negative, B cell-negative, NK cell-negative, due to adenosine deaminase deficiency

Fig. 2. Simiele Chiara Results

Cases Giovanni	Diagnosis	chr	Mutated gene	Variant location	ref	alt	Impact	consequence	Associated phenotype
Case 1	pathogenic	20	SLC52A3	20:761009-761015	CGGGGTC	CGGGTC	High	frameshift_variant	SLC52A3-related BROWN-VIALETTO-VAN LAERE Syndrome
Case 2	healthy	20	CYP24A1	20:54157411-54157415	GAAGC	GAAAGC	High	frameshift_variant	HYPERCALCEMIA INFANTILE 1 (AUTOSOMAL RECESSIVE). Being an autosomal dominant case the child is healthy
Case 3	pathogenic	20	SLC12A5	20:46029385-46029387	GAG	GAAG	High	frameshift_variant	DEVELOPMENTAL AND EPILEPTIC ENCEPHALOPATHY 34
Case 4	pathogenic	20	STK4	20:45000482	C	T	High	stop_gained	COMBINED IMMUNODEFICIENCY due to STK4 deficiency
Case 5	pathogenic	20	RBCK1	20:417801	C	T	High	stop_gained	POLYGLUCOSAN BODY MYOPATHY 1 WITH OR WITHOUT IMMUNODEFICIENCY

Fig. 3. Cenzato Giovanni Results

the confidence that a read is mapped to the correct genomic position. It represents the probability of a mapping error: a high MAPQ (e.g., >30) signifies high confidence, while a score of 0 indicates that the read maps to multiple locations.

The Phred quality score must be around 20 for every call and MAPQ must be as high as possible. The quality control workflow followed a hierarchical approach. Initially, Qualimap was used to perform an individual assessment of each sample, providing high-resolution metrics on alignment integrity and coverage depth. Subsequently, these independent reports were integrated using MultiQC.

This aggregation step allowed for a direct comparison of Phred quality scores and mapping statistics throughout the cohort, ensuring that the variants identified in our clinical cases were derived from consistently high-quality sequencing data. The General Statistics sections of the MultiQC reports (**Fig. 4** and **Fig. 5**), representing the Simiele and Cenzato trios, confirm the high technical quality of both datasets.

For all family members, 100% of genomic regions achieved a coverage depth that exceeded 10X, and 95% surpassed the 30X threshold. Furthermore, alignment efficiency was optimal, with a genome-aligned fraction characterized by an error rate below 0.4%, ensuring a robust baseline for variant calling.

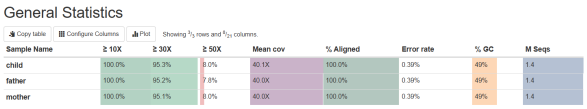


Fig. 4. Screenshot of Chiara Simiele MultiQC Report



Fig. 5. Screenshot of Giovanni Cenzato MultiQC Report

In the MultiQC report, it is also possible to see several quality score plots and an overview of the alignment. Below, two plots have been selected: the mean quality Phred score per position plot in **Figure 6** and the per-sequence quality Phred score plot in **Figure 7**.

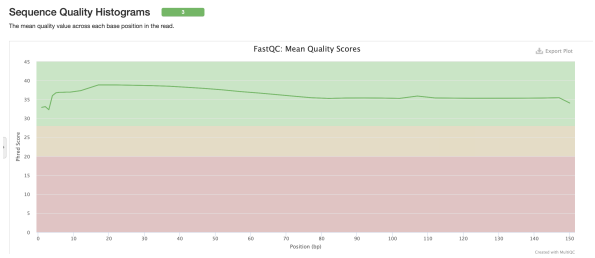


Fig. 6. Screenshot of Simiele's trio MultiQC Report. Quality of the reads along the sequences length

The Mean Quality Scores distribution (**Figure**

6) illustrates the Phred quality score across all base positions (1 to 150 bp) for the first sequenced trio (Mother, Father, and Child). The plot shows a very high quality for all three libraries, which consistently remain within Q > 30 zone. This represents a base call accuracy of at least 99.9%.

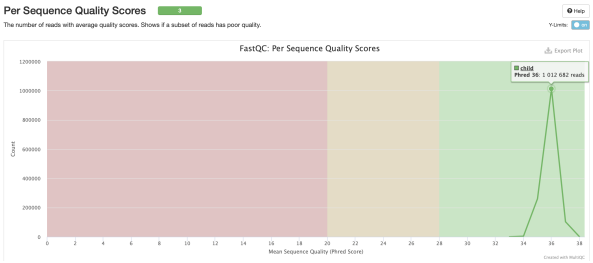


Fig. 7. Screenshot of Simiele's trio MultiQC Report. Per-sequence quality Phred score plot

The Per Sequence Quality Score distribution was analyzed to identify potential subpopulations of low-quality reads within the libraries. This metric represents a histogram of the average Phred scores for every read in the dataset. The plot displays a sharp, unimodal peak centered at a Phred score of approximately 36. This indicates that the vast majority of sequences are of high integrity. Furthermore, over 1,000,000 reads possess a mean Phred score of 36. This signifies that the bulk of the data resides in the "green zone", with negligible representation in the lower-quality regions.

Finally, the locus-specific coverage for the identified variants was validated across all family members using the UCSC Genome Browser. Case 4 from the Cenzato's trio (**Figure 8**) is utilized as a representative model. The visualization confirms that the genomic coordinates harboring the candidate variants were supported by a uniform and substantial read depth in the child and both parents, thereby ensuring high confidence in the clinical relevance of the findings.

In conclusion, the comprehensive suite of quality assessment tools confirmed the high integrity of the raw sequencing data. Consequently, the downstream results and variant calls can be interpreted with a high degree of statistical confidence.

B. Discussion

The results are listed in **Fig 2** and **Fig 3**. Each case number will be followed by the author name for the sake of clarity. Each case is flagged as healthy or pathogenic, depending on the joined evaluation of the filtered vep files generated locally or by using Ensembl VEP Variant Effect Predictor.

The genes identified in the pathogenic cases harbor variants with a high predicted impact on the phenotype and a rare frequency in the population. These protein-coding transcripts are associated with significant functional disruptions, adversely affecting the child's clinical development.

The identified pathogenic variants significantly alter the protein products through several distinct molecular mechanisms:

- Stop gained: this mutation introduces a premature termination codon within the coding sequence. This results in a truncated protein or, more frequently, degradation of the transcript. Given the high impact of the lost C-terminal portion, the resulting protein is rendered non-functional.
- Frameshift variant: this variant mutation involves the insertion or deletion of a number of nucleotides that shift the reading frame. This shift completely alters the product of the transcription from that point onward. In our cases, these variants were caused by a single nucleotide deletion.
- Splice acceptor/donor Variant. These mutations target highly conserved nucleotides at the 5' (donor) or 3' (acceptor) boundaries of an intron. By disrupting these essential recognition sites, the mutation impairs the splicing machinery, leading to aberrant mRNA processing. This typically results in the retention of intronic sequences or the skipping of exons, both of which produce a non-functional mature mRNA.

The Ref and Alt columns represent the reference and alternative nucleotide sequences for each identified variant. In the case of Frameshift Variants, these columns detail the specific insertion or deletion event and the resulting alteration of the open reading frame (ORF).

Case 2 in in **Figure 3** is classified as healthy because the associated phenotype, Infantile Hypercalcemia 1, follows an autosomal recessive inheritance pattern, whereas the inheritance mode observed in this case was autosomal dominant. As the subject carries only a single pathogenic allele, the clinical phenotype is not expected to manifest.

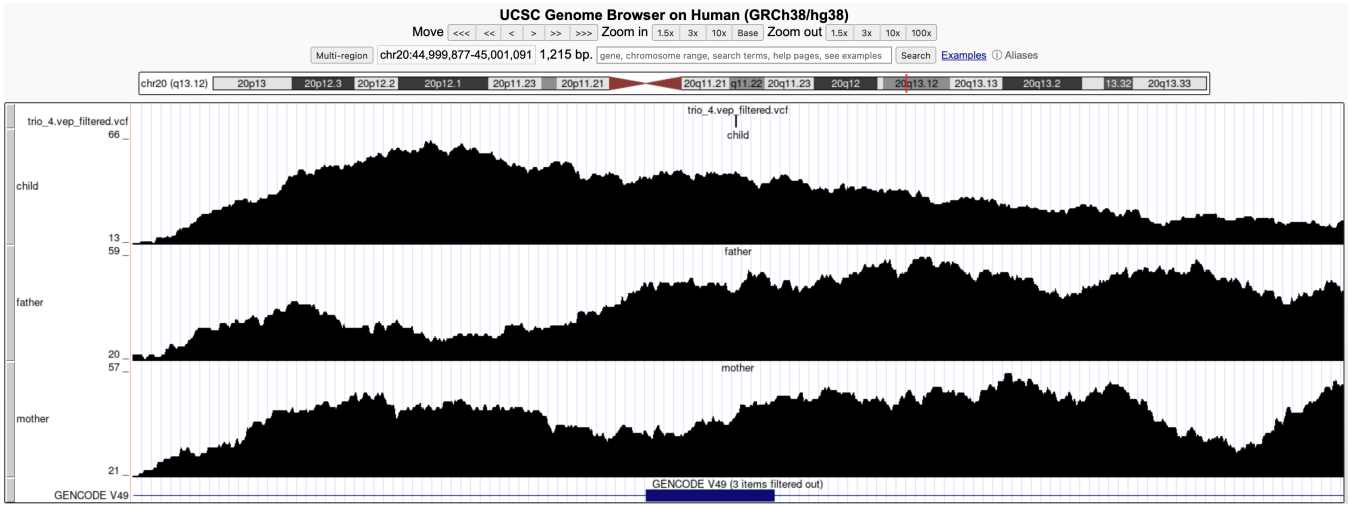


Fig. 8. UCSC Genome Browser screeenshot

4. References

Bedtools: *https://pmc.ncbi.nlm.nih.gov/articles/PMC2832824/*

FastQC: *https://www.bioinformatics.babraham.ac.uk/projects/fastqc/*

QualiMap: *https://pmc.ncbi.nlm.nih.gov/articles/PMC4708105/*

MultiQC: *https://pmc.ncbi.nlm.nih.gov/articles/PMC5039924/*

Samtools: *https://www.htslib.org/doc/1.11/samtools-view.html*

FreeBayes: *https://open.bioqueue.org/home/knowledge/showKnowledge/sig/freebayes*

Bcftools: *https://www.htslib.org/doc/1.0/bcftools.html*

Ensembl Variant Effect Vep Predictor Online: *https://www.ensembl.org/Tools/VEP*

UCSC Genome Browser: *https://genome.ucsc.edu*

Hypercalcemia infantile 1 PubMed: *https://pubmed.ncbi.nlm.nih.gov/38780860/*